

0 5

Scientists investigated the hydrolysis of sucrose in growing plant cells by an enzyme called SPS.

0 5 . 1

Name the products of the hydrolysis of sucrose.

[2 marks]

1 _____

2 _____

0 5 . 2

The scientists grew plant cells in a culture for 12 days. At the start, there were only a few cells in the culture. Each day, they determined the mass of sucrose hydrolysed by SPS in the plant cells in **1 hour**.

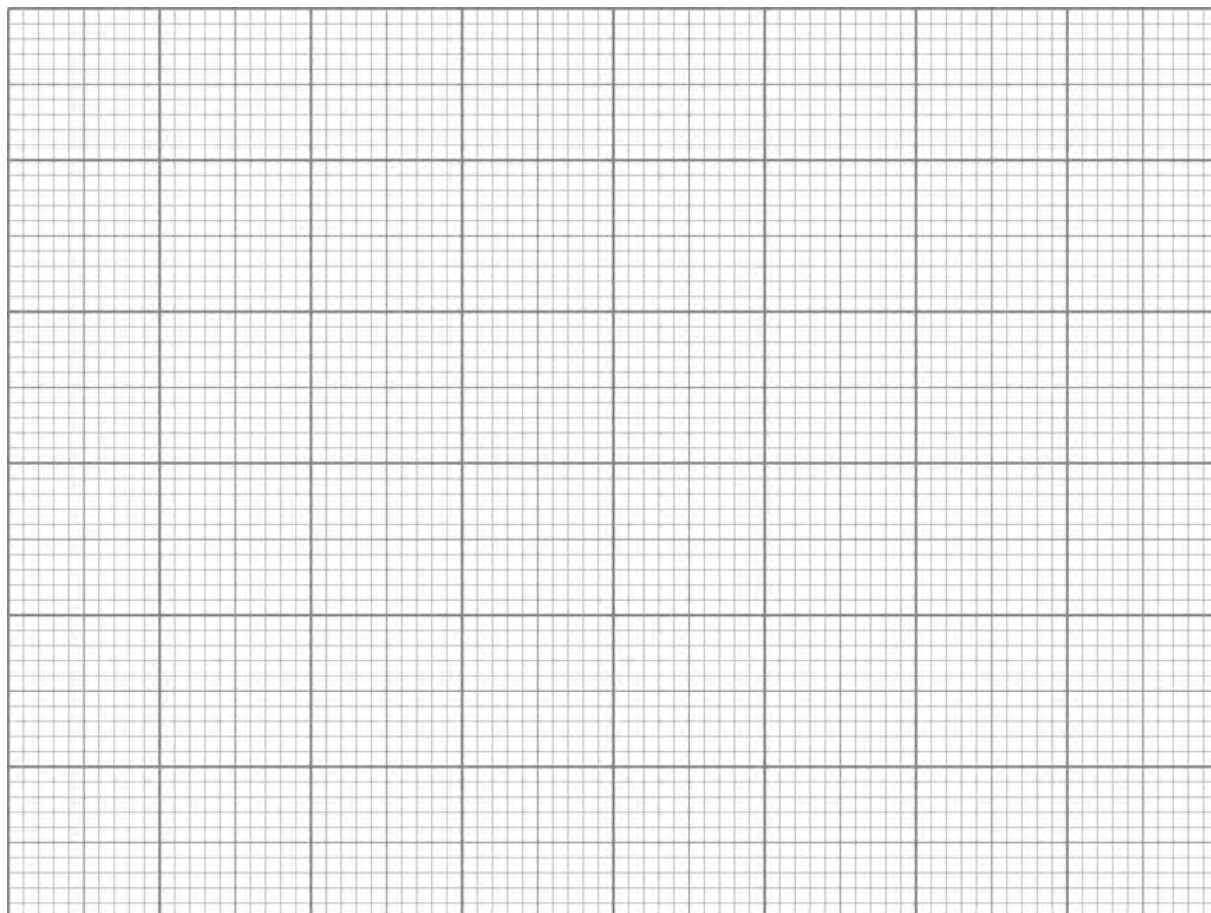
Table 2 shows their results.

Table 2

Day	Mass of sucrose hydrolysed by SPS in 1 hour / μg	Rate of hydrolysis of sucrose by SPS
0	0.07	
2	0.09	
4	0.11	
6	0.15	
8	0.20	
10	0.24	
12	0.24	

For each day, calculate the rate **per minute** of the reaction catalysed by SPS. Record the rates in standard form and plot a suitable graph of your processed data.

[3 marks]

**0 5 . 3**

What can you conclude about the growth of the plant cells from these data?
Explain how you reached your conclusions.

[3 marks]

